Malware Analysis Intro

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Outline

What is Malware Analysis
Types of Malware

Types of Malware Analysis

Create a protected environment

Begin Static Analysis

Definition

Malware Analysis refers to identify the malware, understand how it works and to how to defeat it eventually.

What is malware?

It is a program that can disrupt computer operations, steal sensitive info, gain access to computer systems and spy users.

Key Malware Types

There are many categories of malware out but the most important are:

- Virus
- Trojan
- Worm
- Backdor
- Botnet
- Ransomware
- Spyware
- Keyloggers

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Malware Analysis Types

The are different ways to find and indentify malicious code.

1. **Static Analysis:** It allows us to analyze the malware without executing it. Most of the times the tools we use,help us to find strings,functions or even sumbols inside the malicious code.
2. **Dynamic Analysis:**This method requires to execute the code in order to monitor its activity inside the computer system. We achieve thiw by looking the file system, the process activity, network and system calls.
3. **Memory Analysis:** Memory forensics dives into the volatile memory, aiming to uncover evidence of malware infections and other illicit activities that leave footprints in a system's RAM (Memory Analysis info)

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Before continue

It's vital to build a lab in order to execute or analyze a malware with tools, if you don't want to infect other computer systems and especially your own. In this section, I am going to show you how to create one.
See how its done. I leave a video here: Malware Analysis Lab

Types of Virtual Networks

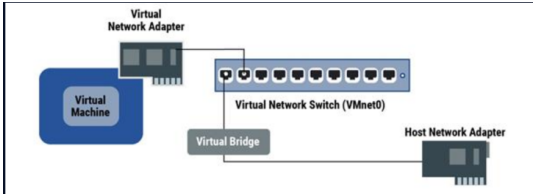

Figure: Bridged Network

A bridged network is a network type where both a virtual machine and the host that it's running on are connected to the same network

Types of Virtual Networks

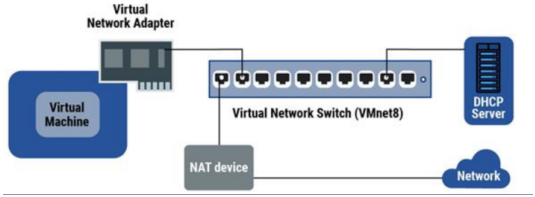

Figure: Nat

On a NAT network, a virtual machine doesn't have its own IP address on the external network. rather, a separate private network is set up on the host computer.

Types of virtual networks

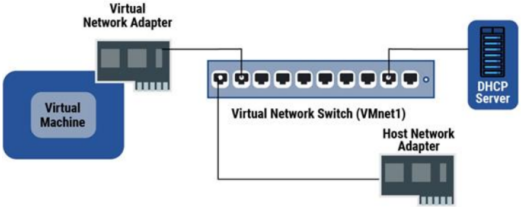

Figure: Host-only

Host-only networking creates a private internal network for the VMs to connect to, analogous to a NAT network. still, without IP address restatement, the VMs can only stay in the private network and don't have direct access to the public external network.

The establishment requires[1]:

1. Creating a **isolated network**: Servers or computer systems are connected in an environment which has no connection with the internet. In your virtual manager(eg. 'Virtual Box','Hyper V'(Windows),'VMWare','QEMU'(Linux only),etc..).
2. Download Kali Linux. It is preferred to download the VM version (KALI)
3. Download images from other operating systems. You can download the edition of your Windows you have or you can experiment with Linux. The choice is yours.

In the internet and the are already images of vulnerable Linux systems.Check Metasploitable

[1]We are going to use Kali to create the lab. Obviously you can create it using REMNux like in the video. REMNux is only for Malware Analysis

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Tools we need

After you create your protected environment you should check if you have available the tools that you need to analyze it:

1. **Radare2** it's a reverse engineering tool(Check if its preinstalled in your Kali).Functions of this programThe Official Radare2 Book
2. **Virus Total** is a tool that given the hash of the malware it checks if it matches with the existing hash in many antivirus tools. It also gives general information on the malware.
3. **MalwareBazaar** it is a site where malware files are stored. However, be careful...

Begin

Firstly you need to download the malware on your Kali machine. For that you need to switch the network from the isolated you build to your default NAT network.

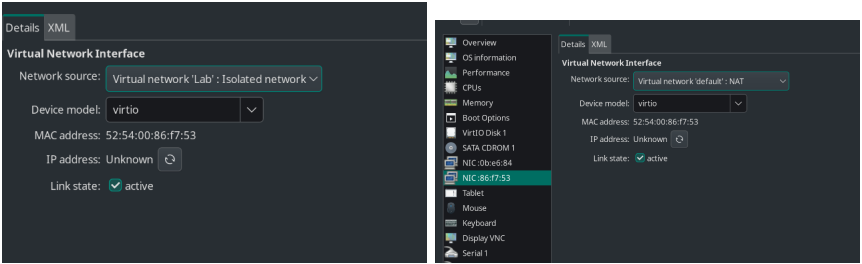

Figure: Change network settings in QEMU

When you download switch to your isolated network.

When you download the zip file from the site and switch the network settings then you excract the file.

Disclaimer!: It will ask you to type the password. Its an information that the site gives you

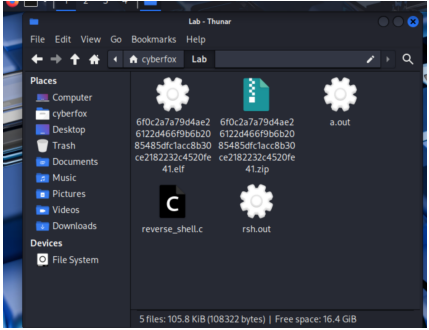

Figure: Change network settings in QEMU

As you can see I extracted the zip file.

First Exercise

For our first exercise we are going to identify the malware.
The first thing that we see from the '6..41.elf' is that the '.elf' file. ELF is short for Executable and Linkable Format. It's a format used for storing binaries, libraries, and core dumps on disks in Linux and Unix-based systems (What is ELF).
In other words it targets Unix based systems. Just be looking in Virus Total we can see:

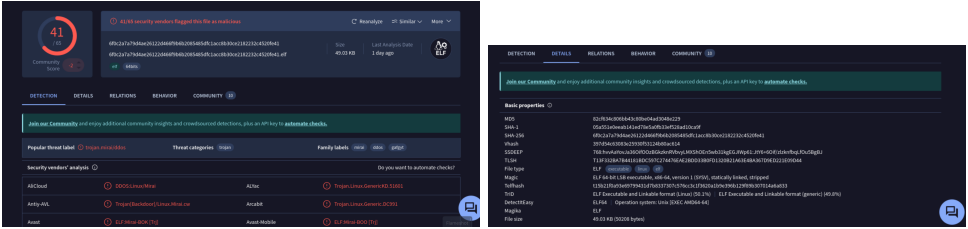

Figure: Looking in Virus Total

First Exercise Summary

What can we say from Virus Total?
Just by looking these 2 pictures we can easily assume:

1. The obvious the file is malicious
2. Is categorized as Trojan, which can occur a DDOS attack and can infected IOT devices [2]
3. Which antivirus systems can identify this file as malicious and how many.

[2] One of the labels is Mirai. Mirai is malware that infects smart devices that run on ARC processors, turning them into a network of remotely controlled bots or "zombies". This network of bots, called a botnet, is often used to launch DDoS attacks.(What is Mirai)

Second Exercise

For our second exercise, we are going to use reverse engineering tools. Specifically, Radare2. For this exercise, it is good to know assembly because the tool is a disassembler. Even if you know arm or risk-v you should understand a great part of the code.
But if you are new to assembly code or you want to get your hands on cisc based architecture assembly you should take a look at: Amd 64 assembly.

Footages of second exercise